

and of duplications. However, these errors can be randomized to have zero expectations, though they introduce errors into census, or additional errors into samples.

Sampling of Homeless People

A major challenge in counting the homeless is choosing which of the many definitions of homeless to use. Does one include only the literal homeless, usually meaning those who reside in homeless shelters or on the street (Glasser et al. 2014). An important study on estimating the homeless population includes the work of Rossi et al. (1987) who sampled various networks within the homeless of Chicago periphery.

Most sample-based studies of the homeless are basically designed to measure characteristics of the currently homeless. The main challenge for designing such samples is that there typically is no fixed reference on which to base a sampling frame (Sumner et al. 2001). Due to several difficulties related to listing of homeless people within a perfect sampling frame, many homeless studies in the past were based on the nonprobability samples. Recently, multistage probability samples have been used for alternative estimation.

Location Sampling Technique

In essence, location sampling is sampling visitors to specified locations. The technique is widely used for two different purposes. One is to sample the population of *visits*, and the other is to sample the population of *visitors* to the locations (Kalton 1991).

There are many hard-to-sample populations for which the probability sampling procedures may not be suitable. Location sampling, also known as *venue-based sampling*, *time-space sampling*, *center sampling*, and *intercept sampling*, is an important additional technique for some of these populations. Many applications of location sampling use nonprobability designs, but the technique can be structured in an attempt to satisfy the requirements of probability sampling (Kalton 2014).

Nonprobability Sampling from Internet Users

Couper (2000) differentiates three types of nonprobability Web surveys that are basically availability samples: *Web surveys as entertainment*, *self-selected Web surveys*, and *volunteer panels of Internet users*. In addition, Ayhan (2005) had proposed a sample weighting adjustment methodology for the nonprobability sample which is selected from voluntary participation Web surveys. Later, Aşan and Ayhan (2013) proposed domain adjustment procedures for nonprobability samples from Internet surveys.

The following points have been emphasized by Weisberg (2005). First, the usual statistical advice is to avoid nonprobability samples. Second, many surveys either explicitly use nonprobability sampling or do so implicitly. Third, the balancing of survey errors and survey costs can sometimes justify nonprobability sampling, regardless of the usual textbook injunctions against it.

About the Author

Dr. Öztaş Ayhan is Professor Emeritus in the Department of Statistics at Middle East Technical University, Ankara, Türkiye. His research interests are survey sampling techniques, survey methodology research, and web survey methodology. During the 1990–1993 period, he served as the Presidential Adviser and Director of the Technical Services Department of the State Institute of Statistics Türkiye. He is an elected extraordinary member of the International Statistical Institute, since 1995. He has been the Country Representative for Türkiye of the International Association of Survey Statisticians, since 1989. Also Vice President of Turkish Statistical Association (2007–2009). Professor Ayhan has published 17 books/monographs and more than 100 research papers in international/national journals and conferences. Professor Ayhan is an active member of the ISI, IASS, IAOS, TSA, TPA, and several other scientific professional bodies.

References

- Aşan, Z., Ayhan, H.Ö.: Sampling frame coverage and domain adjustment procedures for internet surveys. *Qual. Quant.* **47**, 3031–3042 (2013)
- Ayhan, H.Ö.: Sample adjustment weights for internet surveys: restricted access versus voluntary participation. In: Dijkum, C., Blasius, J., Durand, C. (eds.) *Recent Developments and Applications in Social Research Methodology*, pp. 1–8. Barbara Budrich Publishers, Leverkusen-Opladen (2005)
- Ayhan, H.Ö., Ekni, S.: Coverage error in population censuses: the case of Turkey. *Surv. Methodol.* **29**(2), 155–165 (2003)
- Couper, M.P.: Web surveys: a review of issues and approaches. *Public Opin. Q.* **64**, 464–494 (2000)
- Fowler, F.J.: *Survey Research Methods Applied Social Research Methods Series Volume 1*, 3rd edn. Sage Publications, Thousand Oaks (2002)
- Glasser, I., Hirsch, E., Chan, A.: Chap. 9: Reaching and enumerating homeless populations. In: Tourangeau, R., Edwards, B., Johnson, T.P., Wolter, K.M., Bates, N. (eds.) *Hard-to-Survey Populations*, pp. 180–200. Cambridge University Press, Cambridge (2014)
- Goodman, L.A.: Snowball sampling. *Ann. Math. Stat.* **32**(1), 148–170 (1961)
- Kalsbeek, W.D., Cross, A.R.: Problems in sampling nomadic populations. In: *Proceedings of the ASA Survey Research Methods Section*, pp. 398–402 (1982)
- Kalton, G.: *Introduction to Survey Sampling Quantitative Applications in the Social Sciences Series No. 35*. Sage Publications, Thousand Oaks (1983)
- Kalton, G.: Sampling flows of mobile populations. *Surv. Methodol.* **17**, 183–194 (1991)
- Kalton, G.: Practical methods of sampling rare and mobile populations. *Stat. Transit.* **6**, 491–501 (2003)
- Kalton, G.: Chap. 19: Probability sampling methods for hard-to-sample populations. In: Tourangeau, R., Edwards, B., Johnson, T.P., Wolter, K.M., Bates, N. (eds.) *Hard-to-Survey Populations*, pp. 401–423. Cambridge University Press, Cambridge (2014)
- Kish, L.: *Survey Sampling*. Wiley, New York (1965)
- Kish, L.: *Sampling Methods for Agricultural Surveys* *FAO Statistical Development Series No. 3*. Food and Agriculture Organization of the United Nations, Rome (1989)
- Kish, L.: Nomads: Enumerating and sampling. *Questions/Answers (1978–1994) from the Survey Statistician*. International Association of Survey Statisticians, Section of the International Statistical Association, pp. 4–5. (1995)
- Moser, C.A., Kalton, G.: *Survey Methods in Social Investigation*, 2nd edn. Heinemann Educational Books, London (1979)
- Rossi, P., Wright, J., Fisher, G., Willis, G.: The urban homeless: estimating composition and size. *Science*. **235**, 1336–1341 (1987)
- Stephan, F.F., McCarthy, P.J.: *Sampling Opinions*. Wiley, New York (1958)
- Sumner, G.C., Andersen, R.M., Wenzel, S.L., Gelberg, L.: Weighting for period perspective in samples of the homeless. *Am. Behav. Sci.* **45**(1), 80–104 (2001)
- Weisberg, H.F.: *The Total Survey Error Approach: A Guide to the New Science of Survey Research*. The University of Chicago Press, Chicago (2005)
- Yates, F.: *Sampling Methods for Censuses and Surveys*, 3rd edn. Hafner Publishing Company, New York (1960)

Nonresponse in Surveys

Jelke Bethlehem

University of Amsterdam, Amsterdam,
The Netherlands
Statistics Netherlands, The Hague,
The Netherlands

Our complex society experiences an ever growing demand for statistical information relating to social, demographic, industrial, economic, financial, political, and cultural situation of the country. Such information enables policy makers and others to take informed decisions for a better future. Sometimes, statistical information can be retrieved from administrative sources. More often there is a lack of such sources. In this case, the sample survey is a powerful instrument to collect new statistical information.

A sample survey collects information on a small part of the population. In principle, this sample only provides information about the selected elements of the population. Nevertheless, if the sample is selected using a proper sampling design, it is also possible to make inference about the population as a whole.

Many things can go wrong when carrying out a survey. There are all kinds of phenomena that may have a negative impact on the quality of the survey outcomes. Nonresponse is one such problem.

Nonresponse is the phenomenon that elements in the selected sample do not provide the requested information, or that the provided information is unusable. Nonresponse comes in two forms. *Unit nonresponse* denotes the situation in which all requested information